The MRS, the best feasible bundle, utility functions

Lecture notes

What this lecture is about. Lecture 6 described what a consumer *wants* with indifference curves. Today we measure the trade-off along a curve (the marginal rate of substitution), see that different people at the same bundle trade off differently, bring the budget line back to find the best feasible bundle, and finally write preferences as formulas: utility functions.

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1 Indifference curves, briefly

An **indifference curve** joins all bundles a person likes exactly as much as one bundle A. Because more is better, bundles up and to the right of A are better and bundles down and to the left are worse, so the curve slopes down. Curves further out are better, and two curves never cross (transitivity and more is better). With convex preferences the curves bow in toward the origin (or are straight in places: weakly convex).

2 Trade-offs: the marginal rate of substitution

2.1 How much would you give up?

Indifference curves show trade-offs. Start at a bundle and ask: *how much of everything else* (y) *would you give up for one more coffee* (x)*, and still be equally happy?* The answer is the marginal rate of substitution.

Marginal rate of substitution (MRS)

The **MRS** at a bundle is the amount of y the consumer is willing to give up for one more unit of x while staying on the same indifference curve. In the picture it is the steepness of the indifference curve at that bundle: the slope without its minus sign.

The MRS is a *willingness*: it is inside the consumer's head. The relative price of the budget line (lecture 5) is what the *market* asks. Keep the two apart.

Insight: your rate and the market's rate are both Δy over Δx

Every slope is rise over run: $\Delta y/\Delta x$, with y on top and x at the bottom. Both rates answer the same question, *how much y goes for one more x?*

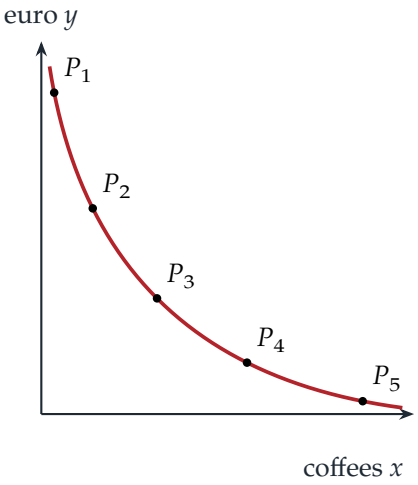
- **The market (budget line).** One more coffee ($\Delta x = +1$) costs p_x euros, so you have p_x/p_y units less of y: $\Delta y = -p_x/p_y$. The slope is $\Delta y/\Delta x = -p_x/p_y$; with $p_y = 1$ it is just $-p_x$.
- **You (indifference curve).** One more coffee is worth some amount of y to you: MRS $= -\Delta y/\Delta x$ along your curve.

Check with numbers: giving up €9 for 3 more coffees means $-\Delta y/\Delta x = 9/3 = 3$ euros per coffee. The upside-down ratio $\Delta x/\Delta y = 1/3$ would be coffees per euro: the same trade-off, but not the MRS.

2.2 Diminishing MRS

Here are five bundles on one indifference curve (coffees, euro of everything else):

bundle	coffees x	euro y	€ per extra coffee
P_1	1	25	
P_2	4	16	$9/3 = 3$
P_3	9	9	$7/5 = 1.4$
P_4	16	4	$5/7 \approx 0.71$
P_5	25	1	$3/9 \approx 0.33$

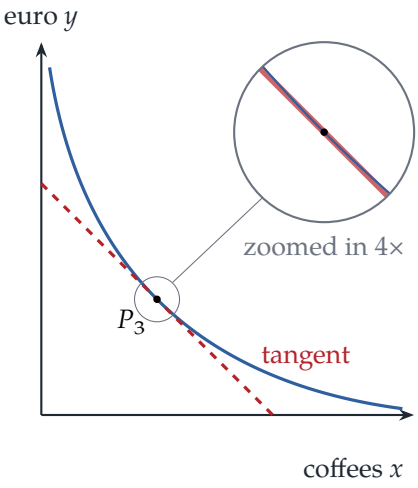


With few coffees and lots of money ($P_1 \to P_2$), you give up €3 for each extra coffee. With many coffees and little money ($P_4 \to P_5$), you give up only about €0.33 for each. The more of a good you already have, the less of the other good you will give up for one more unit of it: your willingness to pay for one more coffee, in euros of everything else, falls. These numbers are averages over each step; the MRS at the points themselves (the tangent slopes, section 2.3) is 5 at P_1, 2 at P_2, 1 at P_3, 0.5 at P_4 and 0.2 at P_5. This is a **diminishing MRS**, and it is exactly what makes the curve bowed toward the origin: steep on the left, flat on the right. For smooth curves, convexity and a diminishing (never rising) MRS are two descriptions of the same shape.

2.3 The MRS at a point: zoom in until the curve is straight

The numbers in the table are averages over a stretch of the curve. The MRS *at* one bundle is the steepness of the curve right there. How do we measure the steepness of a curve? We zoom in.

Look at $P_3 = (9, 9)$ through a magnifying glass. The closer you look, the straighter the curve looks. Very close up, it is almost the same as the straight line that just touches it at P_3: the **tangent line**. The MRS at P_3 is the steepness of that tangent. On this curve the tangent at P_3 has slope -1, so the MRS at P_3 is 1: around 9 coffees and €9, one more coffee is worth about €1 to you.
This is how mathematics deals with curves: replace the curve by its tangent line near the point you care about (**linearisation**). The slope of the tangent is what a derivative measures.



Insight: how we get the slope at one point: shrink the step

Start at $P_3 = (9, 9)$ and take steps of different sizes along the curve. For each step, the trade-off per coffee is $-\Delta y/\Delta x$:

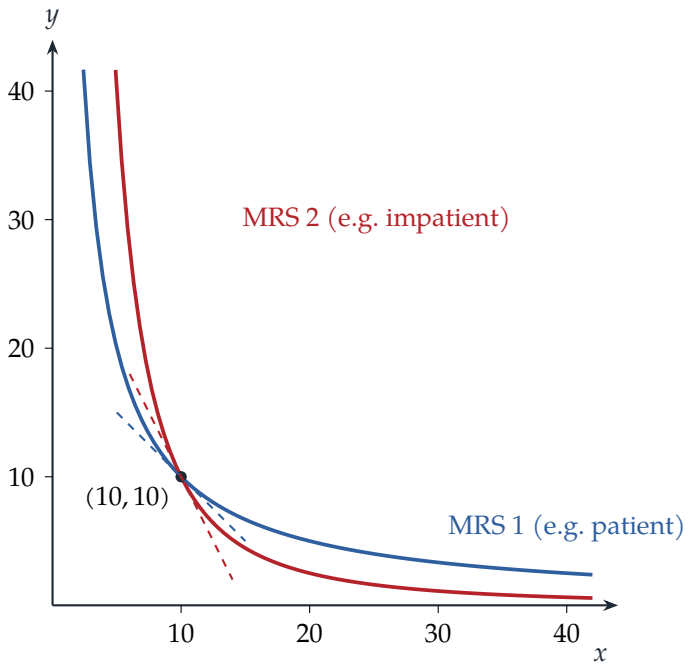
step	Δx	Δy	per coffee, $-\Delta y/\Delta x$
to $(16, 4)$	7	-5	0.71
to $(12, 6.43)$	3	-2.57	0.86
to $(10, 8.05)$	1	-0.95	0.95
to $(9.1, 8.90)$	0.1	-0.099	0.99

As the step shrinks, the number settles at 1. That limit is the slope of the tangent at P_3 (without the minus sign): the MRS at P_3 is 1. Next lecture we get the same number directly from a utility function, with partial derivatives, holding utility constant.

3 Five cases: same bundle, different people

Two people can stand at the *same* bundle and still trade off very differently: their indifference curves through that bundle have different slopes. In every case below the bundle is $(10, 10)$, and the MRS is how much y the person gives up for one more x there.

case (x, y)	type 1	MRS	type 2	MRS
time (€ today, € in one year)	patient	1	impatient	2
sharing (slices for me, slices for my flatmate)	altruistic	1	selfish	2
giving (€ to a local charity, € to a charity abroad)	balanced	1	local first	2
help for two groups (doses for a US group, doses for a China group)	unbiased	1	biased toward the US group	2
work and leisure (hours of leisure, income in €10s)	leisure lover	2	workaholic	$\frac{1}{2}$

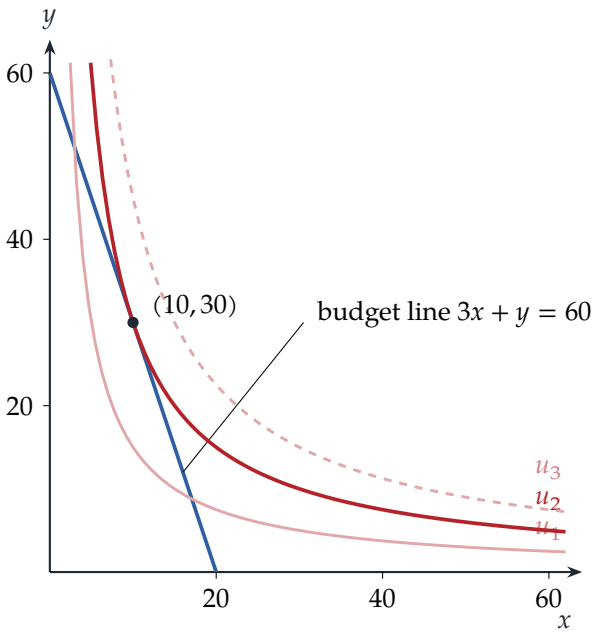


Insight: the slope is the story

The impatient person gives up €2 next year for €1 more today; the patient person only €1. A bias between two groups shows up the same way: as a steeper or flatter curve through the same bundle (a bias toward the other group would give an MRS of $\frac{1}{2}$). A steep curve (high MRS) means that, at this bundle, you value x a lot compared with y; a flat curve (low MRS) means you don't value x that much. The MRS describes *how* a person trades off, not whether the trade-off is good or bad.

4 The best feasible bundle

Now we put what the consumer *can* do (the budget line) and what she *wants* (the indifference curves) in one picture. Take $p_x = 3$, $p_y = 1$ and a budget of €60: the budget line is $3x + y = 60$, from $(0, 60)$ to $(20, 0)$.



- **Feasible** means on or below the budget line. **Best** means on the highest indifference curve that is still feasible.
- u_1 cuts the line (at about $(2.9, 51.2)$ and $(17.1, 8.8)$): moving along the line between those points reaches a better curve, so no point of u_1 is the best.
- u_3 is better but out of reach: none of its bundles is affordable.
- u_2 just touches the line at $(10, 30)$: the **best feasible bundle**.

The best feasible bundle (interior)

Where the best indifference curve touches the budget line, the two have the same slope:

$$MRS = \frac{p_x}{p_y}.$$

Here: the MRS at $(10, 30)$ is 3, and $p_x/p_y = 3/1 = 3$.

Insight: bang for the buck

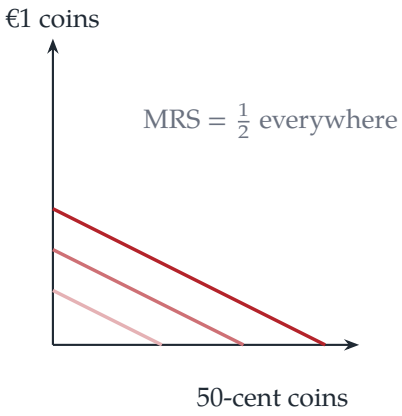
The MRS is what one more x is worth to you, in units of y; the price ratio is what it costs, in units of y. If your MRS is above the price ratio, one more x is worth more than it costs: buy more x. If it is below, buy less x. You stop where they are equal. We only used which curve is higher, never by how much: the argument is *ordinal*.

5 Perfect substitutes and perfect complements

Two extreme kinds of preferences are useful benchmarks. We can describe both with the trade-off question alone, without any formula: *how much y would you give up for one more x?*

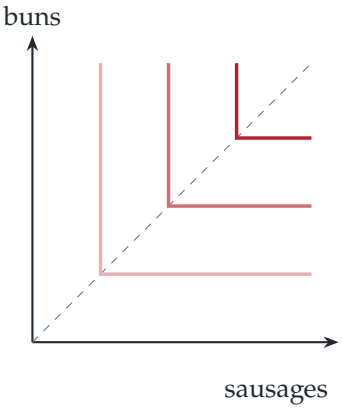
5.1 Perfect substitutes: the answer never changes

Two goods are **perfect substitutes** if they do the same job at a fixed rate. Think of 50-cent coins (x) and €1 coins (y). Two 50-cent coins do exactly what one €1 coin does, and you only care about how much money you have. Then one more 50-cent coin is always worth half a €1 coin to you, whether you have many 50-cent coins or none.
The MRS is **constant**, so the indifference curves are **straight parallel lines**. The rate does not have to be one for one: here it is 2 to 1.

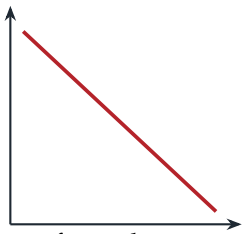


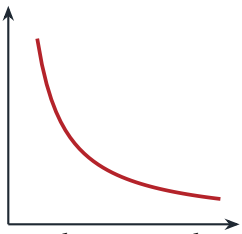
5.2 Perfect complements: no swapping at all

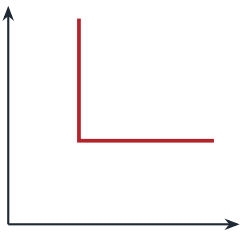
Two goods are **perfect complements** if they are used in fixed proportions. A hot dog needs one sausage and one bun. Five sausages and five buns give you five hot dogs. A sixth sausage without a sixth bun gives you nothing extra; so does a sixth bun without a sixth sausage.
The indifference curves are **L-shaped**, with the corner on the line "as many buns as sausages". On the flat and vertical arms, more of one good alone does not help. Here "more is better" only holds in the weak form: more of *both* is better. On the flat arm the MRS is 0: an extra sausage alone is worth no buns. On the vertical arm the curve is vertical. At the corner the slope jumps, so there is no single MRS there.



5.3 One scale from "swap freely" to "never swap"


perfect substitutes:
straight lines,
swap rate fixed


ordinary goods:
bowed-in curves,
swap rate falls


perfect complements:
L-shapes, no swapping

Both extremes are only *weakly* convex: a mix of two equally good bundles is always at least as good, but for some pairs it is only *equally* good. For substitutes that is any two points on the same line; for complements, any two points on the same arm of the L. So straight lines and L-shapes do not violate convexity; they fail only *strict* convexity.

Insight: how to tell them apart without formulas

Ask the trade-off question at several bundles. If the answer is "always the same amount", the goods are perfect substitutes and the curves are straight lines. If the answer is "nothing, unless I also get more of the other good", they are perfect complements and the curves are L-shapes. Everything else is in between: the answer depends on how much you already have, and the curves bow in.

6 Utility functions

Utility function

A **utility function** $u(x, y)$ gives every bundle a number, so that a better bundle gets a higher number and equally good bundles get the same number. An indifference curve is all bundles with the same number: $u(x, y) = 10$, $u(x, y) = 20$, and so on.

6.1 Ordinal: only the order counts

Utility

A **utility function** gives each bundle a number, so that a better bundle gets a higher number: $u(A) > u(B)$ exactly when $A \succ B$, and $u(A) = u(B)$ exactly when $A \sim B$.

Take three bundles at the same number of coffees, one on each of three indifference curves: $A = (10, 6)$, $B = (10, 12)$ and $C = (10, 24)$. More euros, so $C \succ B \succ A$. Now label them with three different utility functions:

bundle	u	$2u + 5$	u^2
A	1	7	1
B	2	9	4
C	3	11	9
order	$A < B < C$	$A < B < C$	$A < B < C$
$C \div A$	3	≈ 1.6	9

- **Ordinal:** every column gives the same order, so every column describes the same preferences and the same indifference curves. (Squaring keeps the order here only because all the u values are positive.) Only the order carries information.

- **Cardinal** would mean that the numbers themselves say something, such as "C is three times as good as A". But that answer is 3, about 1.6 or 9, depending only on which labels we picked. So we do not read utility numbers that way.

6.2 The standard formulas

- **Perfect substitutes:** $u(x, y) = a\,x + b\,y$. The curves are straight lines, and the MRS is $\frac{a}{b}$ at every bundle. Example: two brands of bottled water, $u = x + y$ (a litre is a litre).
- **Perfect complements:** $u(x, y) = \min\{a\,x,\ b\,y\}$. The curves are L-shapes with the corner where $a\,x = b\,y$. Example: sausages and buns, $u = \min\{x, y\}$ hot dogs; 10 sausages and 15 buns still make 10 hot dogs.
- **Cobb-Douglas:** $u(x, y) = x^{\alpha} y^{\beta}$. Smooth, bowed-in curves; the MRS falls along the curve. The curves of our five cases are of this kind: $x\,y$ (MRS 1 at $(10, 10)$), $x^2 y$ (MRS 2) and $x\,y^2$ (MRS $\frac{1}{2}$). A bigger α makes x matter more: a steeper curve through the same bundle.
- **Mixtures:** for example $u(x, y) = \alpha\,x + y^{\beta}$ with $\beta < 1$. In x every unit adds the same, as with perfect substitutes; in y each extra unit adds a little less. Example: $u = 2x + \sqrt{y}$. Its curves $y = (u - 2x)^2$ bow in, and each is the previous one shifted sideways (for $u = 8, 10, 12$ they meet the x-axis at 4, 5 and 6), so they can reach the axis: corners are possible. The curves are neither straight lines nor L-shapes.

7 What choices reveal

7.1 From choices to preferences

Choices are not preferences. A *preference* is what you like; it is inside you and nobody can see it. A *choice* is what you actually pick, and it depends on two things: what you like *and* what you can afford. In everyday language, and in many languages, the two words are used almost interchangeably ("my choice", "my preference"). In economics we keep them apart, because the budget sits between them: the same preferences lead to different choices when prices or income change.

So far we assumed that we know a consumer's preferences. In practice we only see what people *choose*. Can we check the four rules from choices alone? Yes, and this is a large part of modern experimental economics.

In a budget experiment (Choi, Fisman, Gale and Kariv, 2007, and many studies since), each participant sees a series of straight budget lines, one at a time, and picks one point on each line. The lines are drawn at random and cross each other often. The researcher sees nothing but the lines and the chosen points.

It helps to write every line in the intercept form of lecture 5,

$$\frac{x}{X} + \frac{y}{Y} = 1,$$

where X and Y are the two ends. The left-hand side is then the **cost of a bundle as a share of the budget**: a bundle with $\frac{x}{X} + \frac{y}{Y} < 1$ is affordable on that line with money to spare.

Directly revealed preferred

If a consumer chose bundle A on a line where bundle B was also affordable, then A is **directly revealed preferred** to B. She could have had B and took A. If B was strictly cheaper than the budget, A is revealed *strictly* preferred: with more is better, she could have had B plus some extra money.

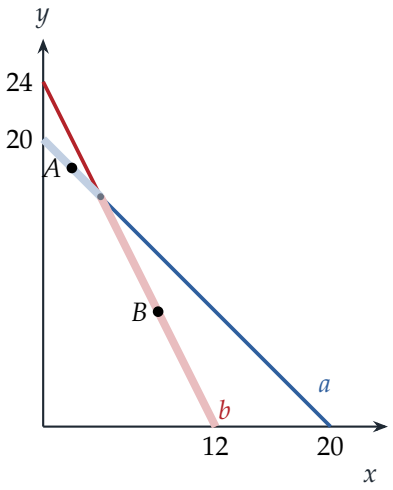
7.2 Two lines: rational or not

Take line a with ends $(20, 20)$, that is $\frac{x}{20} + \frac{y}{20} = 1$, and line b with ends $(12, 24)$. The two lines cross at $(4, 16)$.

A contradiction. On line a she picks $A = (2, 18)$. On line b she picks $B = (8, 8)$.

- Was B affordable on line a? Its cost share is $\frac{8}{20} + \frac{8}{20} = 0.8 < 1$. Yes, so A is revealed preferred to B.
- Was A affordable on line b? $\frac{2}{12} + \frac{18}{24} \approx 0.92 < 1$. Yes, so B is revealed preferred to A.

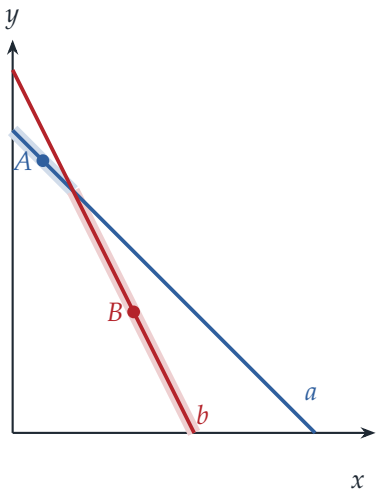
Each choice was affordable when the other was made. No consumer with preferences that follow our rules chooses like this. This is a violation of the **weak axiom of revealed preference** (WARP). It can only happen if both choices sit on the "inner" pieces of the two lines, the pieces that lie below the other line.



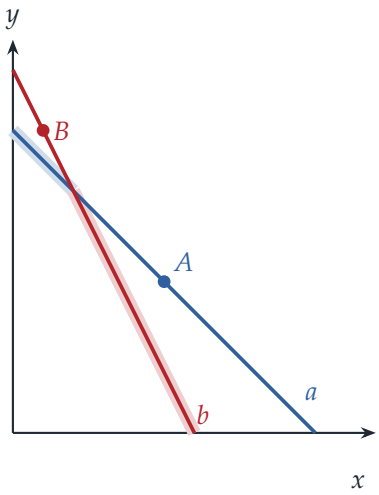
shaded: the inner pieces

7.3 All four cases on the same two lines

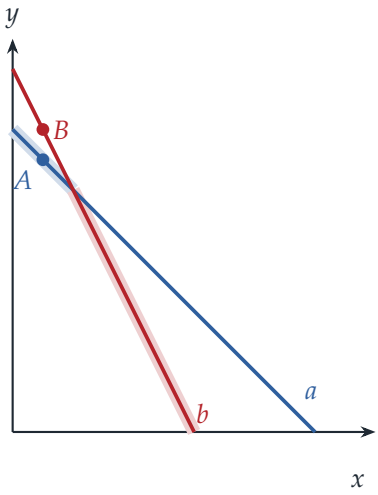
Whether two choices contradict each other depends only on *where* on the two lines they sit. Each line has an **inner piece** (the part below the other line, shaded) and an **outer piece** (the part above the other line). A bundle on an inner piece is affordable on the other line; a bundle on an outer piece is not. That gives four cases.



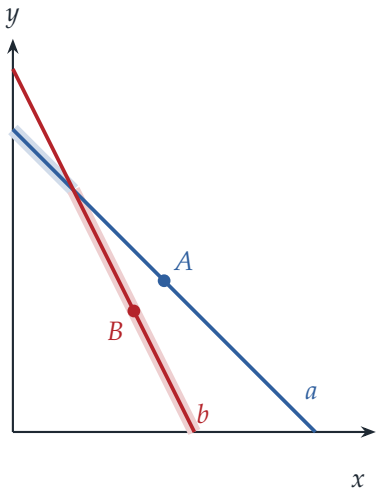
1 · both inner. Each was affordable when the other was picked: a contradiction.



2 · both outer. Neither was affordable on the other line. Nothing is revealed: rational.



3 · A inner, B outer. A was affordable on line b, so B is revealed preferred to A, and nothing the other way: rational.



4 · A outer, B inner. B was affordable on line a, so A is revealed preferred to B, and nothing the other way: rational.

shaded: the inner pieces (the part of each line that lies below the other line)

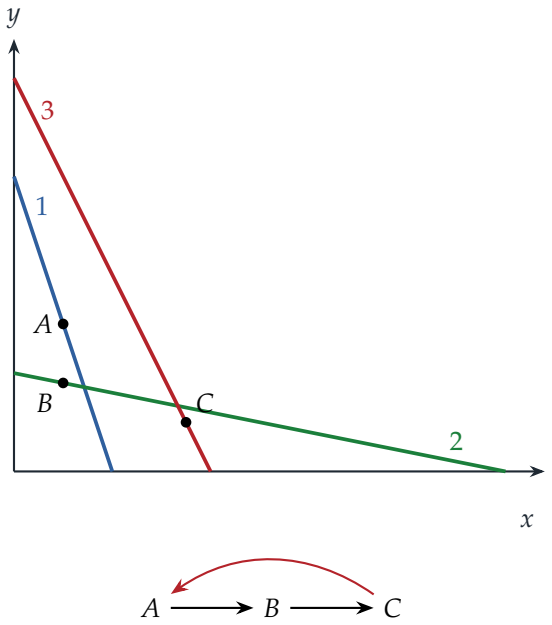
case	A on line a	B on line b	B affordable on a?	A affordable on b?	verdict
1	$(2, 18)$	$(8, 8)$	$0.8 < 1$: yes	$0.92 < 1$: yes	contradiction
2	$(10, 10)$	$(2, 20)$	$1.1 > 1$: no	$1.25 > 1$: no	rational
3	$(2, 18)$	$(2, 20)$	$1.1 > 1$: no	$0.92 < 1$: yes	rational
4	$(10, 10)$	$(8, 8)$	$0.8 < 1$: yes	$1.25 > 1$: no	rational

The numbers are the **cost shares**: $\frac{x}{20} + \frac{y}{20}$ on line a and $\frac{x}{12} + \frac{y}{24}$ on line b. Only case 1 breaks the rules, and only because *both* picks sit on the inner pieces. So the rule you can see on a graph is: **two choices contradict when each lies below the other line** (or when one sits exactly where the lines cross and the other below).

7.4 Three lines: a cycle

With more lines, revealed preferences can be chained: if A is revealed preferred to B and B to C, then A is *indirectly* revealed preferred to C. This is transitivity again. A **cycle** (A over B over C over A) means the choices cannot come from any transitive ranking. The **generalised axiom of revealed preference** (GARP) says: no such cycles with a strict step.

line	ends (X, Y)	choice
1	$(20, 60)$	$A = (10, 30)$
2	$(100, 20)$	$B = (10, 18)$
3	$(40, 80)$	$C = (35, 10)$



- On line 1, B costs $\frac{10}{20} + \frac{18}{60} = 0.8$ of the budget: A revealed preferred to B.
- On line 2, C costs $\frac{35}{100} + \frac{10}{20} = 0.85$: B revealed preferred to C.
- On line 3, A costs $\frac{10}{40} + \frac{30}{80} = 0.625$: C revealed preferred to A.
- Also on line 3, B costs $\frac{10}{40} + \frac{18}{80} \approx 0.48$: C revealed preferred to B as well.

Insight: why A and B alone are fine, but the chain is not

Take only lines 1 and 2. B was affordable on line 1 (cost share 0.8), so A is revealed preferred to B. But A was *not* affordable on line 2: its cost share there is $\frac{10}{100} + \frac{30}{20} = 1.6 > 1$. So there is no "$B$ over A" and no contradiction: this is case 4 of section 7.3 (A on the outer piece, B on the inner piece).

The point $K \approx (14.3, 17.1)$ where lines 1 and 2 cross is on both lines, so it was affordable both times: A is revealed at least as good as K, and B is revealed at least as good as K. That is no problem either: two bundles that both beat K say nothing about each other. (This is only true while we look at lines 1 and 2. Once line 3 comes in, A and B clash too, through C.)

A contradiction needs one bundle over another *and* the other way round. Three lines can deliver that through a chain: A over B (line 1), B over C (line 2), C over A (line 3), so A is over C and C is over A at the same time. (Line 3 also puts C over B, which clashes directly with line 2.)

Here is every check at once: on each line, the cost share of the two bundles that were *not* chosen there.

line (choice)	bundle	cost share	affordable?
1 (picked A)	$B = (10, 18)$	$\frac{10}{20} + \frac{18}{60} = 0.8$	yes: A over B
	$C = (35, 10)$	$\frac{35}{20} + \frac{10}{60} \approx 1.92$	no
2 (picked B)	$A = (10, 30)$	$\frac{10}{100} + \frac{30}{20} = 1.6$	no
	$C = (35, 10)$	$\frac{35}{100} + \frac{10}{20} = 0.85$	yes: B over C
3 (picked C)	$A = (10, 30)$	$\frac{10}{40} + \frac{30}{80} = 0.625$	yes: C over A
	$B = (10, 18)$	$\frac{10}{40} + \frac{18}{80} \approx 0.48$	yes: C over B

So these three choices contain **three contradictions, one for every pair**:

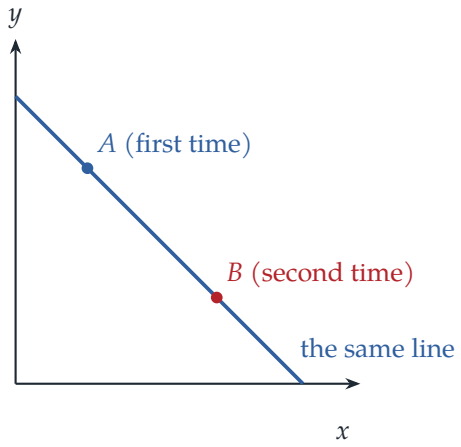
1. B **and** C**, directly.** B over C (line 2) and C over B (line 3): each was affordable when the other was picked. This is case 1 of section 7.3, on two lines.
2. A **and** C**, through the chain.** C over A comes straight from line 3. A over C is only *indirect*: A over B (line 1) and B over C (line 2). C was never affordable when A was picked (its cost share on line 1 is $\frac{35}{20} + \frac{10}{60} \approx 1.92$). The cycle $A \to B \to C \to A$ only shows when you follow the chain.
3. A **and** B**, through the chain.** A over B is direct (line 1, cost share 0.8). B over A is indirect: B over C (line 2) and C over A (line 3).

In a cycle, every pair contradicts: here one directly (B and C), the other two through the chain. Any one of them means the choices violate GARP. With two goods, as here, a cycle always contains a pair of lines that contradict each other directly (here B and C). Cycles in which no pair contradicts directly need three or more goods.

7.5 The same line twice

What if you face the *same* budget line twice and pick two different bundles, A the first time and B the second? Both cost exactly the budget, so each was affordable when the other was picked: A is revealed at least as good as B, and B at least as good as A. Together that says $A \sim B$: you are **indifferent** between them. That is allowed. GARP only forbids a cycle with a *strict* step (one bundle that was strictly cheaper than the budget), and here there is none.

So two different answers on the same line are **not** a contradiction under GARP: they show a tie, for example when your indifference curve runs along the budget line, as with perfect substitutes. They *are* a contradiction if we also assume that the best bundle is unique (strictly convex preferences, where only one point on the line touches the highest curve). Some studies use that stricter test (the strong axiom, SARP).
It becomes a clear violation as soon as one of the two picks is strictly inside the other's budget, which is exactly the two-line case 1 of section 7.3.



both cost the whole budget: a tie, not a cycle

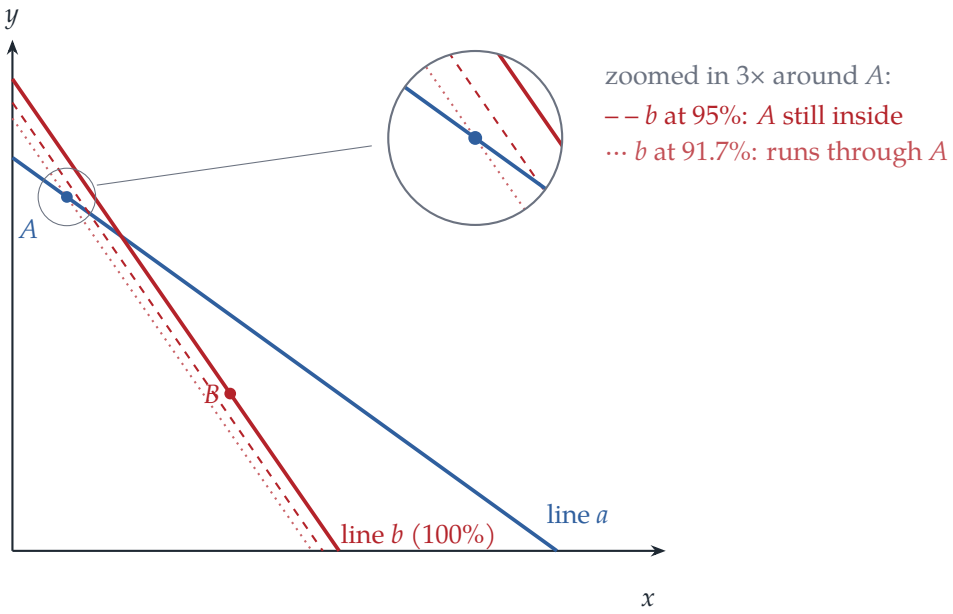
7.6 How close to rational: the efficiency index in a picture

Go back to the contradiction of section 7.3, case 1: $A = (2, 18)$ on line a and $B = (8, 8)$ on line b. On line b you picked B, although A cost only 0.917 of that budget ($\frac{2}{12} + \frac{18}{24}$). So you paid the whole budget for B when A was available for 91.7% of it.

Now **shift every budget line inwards** (burn budget), to a fraction e of its money, and count a bundle as "revealed worse" only if it fits inside the shifted line.

- At $e = 0.95$: line b shifts inwards to ends $(11.4, 22.8)$. A (cost share 0.917) is still inside: still "B over A", still a contradiction.
- At $e = 0.917$: the shifted line b runs exactly through A. Just below that, A is outside: "B over A" disappears, and the contradiction with it.

The share e where the contradictions stop (at every share below it, none is left) is the **critical cost efficiency index**: here CCEI ≈ 0.92. In words: your choices could come from rational preferences for which more is better once we forgive wasting just over 8% of a budget. A fully rational chooser has CCEI = 1; the further below 1, the more money the contradiction "costs".



7.7 Afriat's theorem and how close to rational

Afriat's theorem (1967), in words

A set of choices on budget lines passes GARP **if and only if** there is a well-behaved preference ordering (complete, transitive, more is better, and convex) under which every choice was the best affordable bundle.

This is a striking result. We never need to guess a formula for preferences: checking the choices for cycles is enough. It also means that convexity is an assumption we can never catch failing by watching choices on budget lines. Someone who loves extremes picks an end of the budget line, but so can someone with ordinary bowed-in curves who simply likes that good a lot: all we see is the picked point, so the two look the same. In general, whatever bundles a person picks, as long as the picks contain no cycle, we can always draw convex indifference curves (bowed in, possibly with some straight pieces) under which every pick is a best bundle on its line. So convexity is something we add because it makes the analysis simple, not something budget data could ever reject.

Real people are rarely perfectly rational, so researchers also measure *how close* choices come, with the efficiency index (CCEI) drawn in section 7.6, the efficiency index in a picture: 1 means fully rational, 0.95 means the violations are worth just over 5% of a budget. A useful benchmark: people who click on random points of many crossing lines almost always violate GARP, while most participants in budget experiments come close to full rationality.

8 Next lecture

From a utility function to the MRS by taking derivatives; setting the MRS equal to the price ratio and solving for the best bundle; and the edge cases: corners (perfect substitutes) and

kinks (perfect complements). Then practice.

Summary

- The MRS is how much y you would give up for one more x: the steepness of the indifference curve at a bundle. It falls along a bowed-in curve.
- People at the same bundle can have different MRS: patient vs impatient, altruistic vs selfish, unbiased vs biased.
- The best feasible bundle is on the highest reachable curve; in the interior, $MRS = p_x/p_y$ there.
- Perfect substitutes: straight lines, constant MRS. Perfect complements: L-shapes, fixed proportions.
- A utility function puts numbers on bundles; only the order of the numbers matters (ordinal).

Further reading. Frank, *Microeconomics and Behavior*, pp. 68–71 and Appendix 3A, pp. 83–89 (week 5 page, additional readings; the password is given in class).